

Neutrino Dirac→Majorana reversal — full enumeration + owner assignments (Grace, sweep #1)

Grace | strengthening round 2, item #1 (blocks outreach) | 2026-07-18. I own the full-list cross-ref (per K740). Enumerated the corpus: 68 files mention the old Dirac/ $0\nu\beta\beta$ -forbidden thread; classified into 40 current-facing (REVISE) + 28 historical/adjudication (LEAVE — they record the debate). Data layer swept by me (done). Owners assigned for the rest per K740.

The revision (apply everywhere the OLD thread appears)

- OLD (stale): Dirac neutrinos; S^1 winding / Hopf / $\mathbb{Z}_3$ makes $B-L$ exactly conserved; $0\nu\beta\beta$ forbidden, $|m_{\beta\beta}|=0$; a detection kills BST.
- NEW (current, 2026-07-16, F413/K673, further supported by F582 ν_R = gauge-blind singlet): Majorana neutrinos; the Majorana mass is the $\Delta L=2$ topology-changing term ($B-L$ violated by 2); $0\nu\beta\beta$ OCCURS, $m_{\beta\beta} \in [1.4, 3.7]$ meV (normal ordering, $m_{\nu 1}=0$); a detection SUPPORTS BST, a null below ~ 1 meV falsifies.

☐ DATA LAYER — swept by Grace (DONE, backups made)

- data/bst_predictions.json — already correct (Majorana; this is the one file that propagated 07-16). ☐
- data/bst_particles.json — FIXED (line 171 key_insight: Dirac→Majorana). ☐
- data/bst_seed.md — FIXED (line 148: “Dirac neutrinos” → Majorana + $0\nu\beta\beta$ note). ☐
- data/bst_geometric_invariants.json — FIXED (nu_Dirac→nu_Majorana entry; $|m_{\beta\beta}|$ entry $0 \rightarrow [1.4, 3.7]$ meV). ☐

CURRENT-FACING — REVISE (40 total, by owner)

→ LYRA (Working_Paper + Curriculum + Lyra-authored notes) — physics prose

Working_Paper (referee-facing, HIGHEST priority per K740): - Working_Paper/Vol5_Predictions/Ch01_Predictions_Program (HIGHEST — the predictions volume; falsifier table backwards) - Working_Paper/Vol6_Frontier/Ch01_Deep_Results.md (cleanest-falsifier line states the reverse) - Working_Paper/Vol3_Physics/Ch03_Forces_Cosmology.md - Working_Paper/Vol1_Journey/Ch01_The_Journey.md Lyra-authored notes/papers: - notes/Lyra_T1_Falsifier_Lepton_Count_v0 Lyra_T1_Falsifier_v0_2_CD_Sharpener.md - notes/Lyra_KTypes_Are_Particles_Falsifier_Design_v0_1.md, Lyra_Quasi_Eigentone_Framework_v0_1.md - notes/Lyra_Spectral_Score_Theoretical_Layer_v0_1.md, Lyra_SSG15_Lambda_Coupled_Neutrino_Framework_v0_1.md - notes/Paper_P6_Periodic_Table_Substrate_SM_v0_1/v0_2/ (P6 drafts, Grace+Lyra co-authored)

→ KEEPER (the two source papers, per K740)

- notes/BST_NeutrinolessDoubleBeta.md — whole paper is Dirac; rewrite to the Majorana prediction.
- notes/BST_Conservation_Laws.md — the ORIGIN of “ $B-L$ exact via Hopf → Dirac”; reframe to $\Delta L=2$ -violated + topological-conservation.

→ GRACE (prediction/outreach/falsifier notes + registry — my remaining sweep)

- notes/BST_Neutrino_Predictions.md, BST_Testable_Predictions_Catalog.md, BST_T1158_Falsifiable_Predictions
- notes/BST_Outreach_OnePage_v0.1.md, BST_Outreach_OnePage_v0.2.md (outreach-facing — high priority)
- notes/BST_Preregister_Falsifier_Draft.md, BST_Paper48_What_BST_Forbinds_Draft.md, BST_Paper64_Experiments
- notes/BST_Tests_1.md, BST_Constants_Gap_Audit.md, BST_Interstasis_I20_ParticlePersistence.md
- notes/BST_Odd_Dimensionality_Unification_2026-07-14.md, BST_Substrate_Engineering_Reference_Manual_v0_1
- notes/BST_AC_Theorem_Registry.md, BST_AC_Theorems.md (T1436/T1929/T1945 neutrino-nature rows)

- notes/CI_BOARD.md, notes/.running/RUNNING_NOTES.md, notes/referee_objections_log.md, Paper83_Outline.md
- notes/Cal_Proposal_KTypes_Are_Particles_Falsifier_Design_v0_1.md, notes/maybe/Grace_Cosmology_Substrate

HISTORICAL / ADJUDICATION — LEAVE (28 — they RECORD the debate, revising them erases the audit trail)

Old MESSAGES (04-, 06-), the adjudication F/K-notes (K673, F144, F146, F374, F396, F456, F512, F513, K378, K421, K583, K684), the HonestState ledgers, Investigation boards, and the June-dated grace_/keeper_ notes. These correctly show the Dirac→Majorana decision *as history*; do NOT rewrite.

Λ over-determination retraction (also mine, per K740)

- My memory MEMORY.md carries “ $280 = 2^N \cdot n \cdot C \cdot g$ (5-fold over-determined)” — filing a correcting memory (the “5-fold” is one factorization $\times 5$: $2^3 = 2^N \cdot n = \text{rank}^3 = 8$; Λ is STRUCTURAL, not derived, per K741).
- Cosmology notes carrying the over-determination-DERIVED headline: notes/BST_Paper135_..._Lambda_Casimir, Lyra_L5_v0_3_Elie280, Lyra_F12_..._Lambda; Curriculum Vol4 Ch4 (Lyra). Revision: “280 structural (right form/size, not target-innocent-forced); over-determination retired; needs a mechanism.”

Process gate (per K740, added)

Reversal-propagation checklist: when a prediction flips (07-16 Majorana; K739 $\sin^2 \theta_W$ 3/13), the reversal MUST propagate to papers + curriculum + notes, not just the data layer. This staleness lived 2 days because there was no gate. Added to EOD discipline.

— Grace, 2026-07-18. Full enumeration: 68 files → 40 revise (data ☐ done; Working_Paper→Lyra; 2 papers→Keeper; prediction/outreach notes→Grace) + 28 historical (leave). Standard OLD→NEW revision text above. Blocks outreach until the 40 land.